

1.2 Renewable and Non-Renewable Energy Resources

Energy either exists in the earth or, comes from the outer space. The energy existing in the earth is called capital energy and that comes from the outer space is called celestial or income energy. The capital energy sources are mainly, fossil fuels, nuclear fuels and heat traps. The celestial energy sources include electromagnet, gravitational and particle energy from stars, plants and moon as well as the potential energy of meteorites entering the earth's atmosphere. The useful celestial energy sources are the electromagnetic energy of the earth's sun, called direct solar energy, and the gravitational energy of the earth's moon, producing tidal flows. The other energy sources such as wind, hydel, geothermal, biofuel, etc are derived from the direct solar energy. Due to ever increasing rate of consumption of energy day by day, very rapid depletion of energy is directing us to look at the alternate energy sources such as solar, wind, tidal, ocean geothermal, ocean etc.

The various energy sources are grouped under two main types. They are : Renewable energy sources and Non-Renewable energy sources.

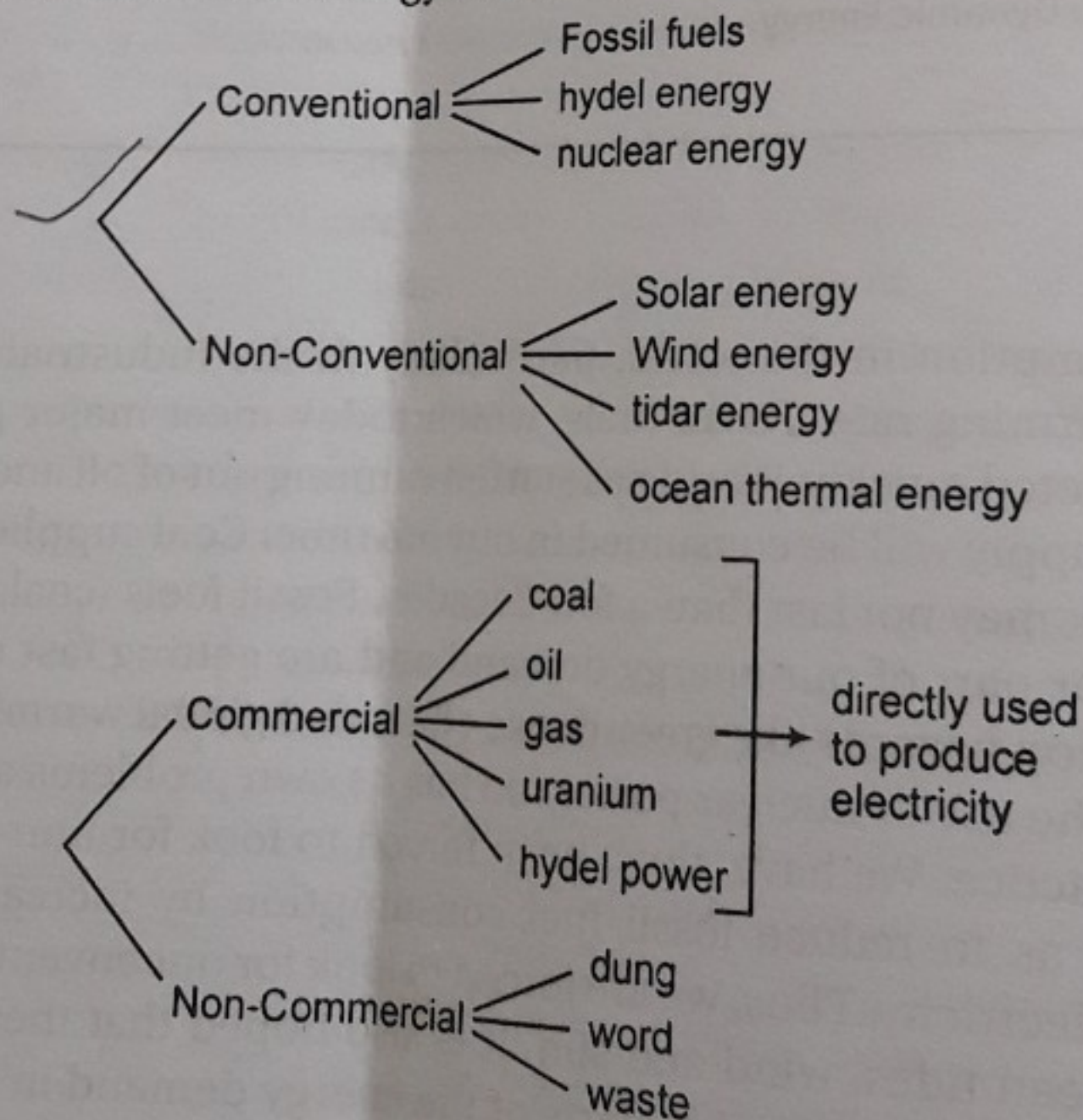
The renewable energy resources are defined as (the energy sources which are produced continuously in nature and are essentially inexhaustible eventually in future.) Examples are:

- (i) Solar energy
- (ii) Wind energy
- (iii) Tidal energy
- (iv) Ocean energy
- (v) Geothermal energy

The non-renewable energy resources are defined as (the energy resources which have been accumulated over the ages and not quickly replaceable when they are exhausted.) Non-renewable sources of energy will get exhausted eventually in future. Examples are:

- (i) fossil fuels
- (ii) nuclear fuels
- (iii) heat traps.

Other ways of classification energy sources :



1.3 Advantages and Disadvantages of Renewable Resources

1.3.1 Advantages

- (i) The renewable energy resources are non-exhaustible
- (ii) Renewable energy is an indigenous resource available in considerable quantities to all developing nations.
- (iii) Several renewable options are financially and economically competitive for certain application, such as in remote locations, where the cost of transmitting electrical power are high.
- (iv) The renewable energy resources can be matched in scale to the need and also they can deliver the energy of the quality that is required for a specific task.
- (v) The diversity of renewable energy resources and their technologies offer more flexibility while designing the conversion systems compare to the conventional energy system.
- (vi) The local or regional self sufficiency in the energy requirement can be achieved either fully or, partially by harnessing the locally available renewable energy which otherwise would be left unutilized.
- (vii) Except the biomass energy source, all other renewable resources offer pollution free environment and also help in maintaining the ecological balance.

1.3.2 Disadvantages

- (i) The intermittent nature of the availability of the energy from the renewable energy resources is a major set back in the continuous supply of energy.
- (ii) Although the supply of energy from the sun is limitless, there is a definite limit to the rate at which the solar energy is received at the earth as it is dependent on the local atmospheric conditions, time of the day, part of the year and also on the latitude of the place.
- (iii) Some of the renewable energy resources such as wind, tidal etc although available in large quantities, they are concentrated only in certain region.
- (iv) The state of the art in harnessing the renewable energy resources is not yet fully developed to meet the present day energy requirements.
- (v) Some of the renewable energy conversion systems such as solar cells, automatic tracking systems for solar concentrators etc, require advanced technologies, hence costlier.
- (vi) The application of renewable energy resources to transport sector has been found to be not viable as on today.

1.4 Forms and Conversions of Renewable Energy

The various forms of the renewable energy sources are solar, wind, tidal, hydro, wave, biomass and geothermal. The energy available from each of these sources can be used to produce mechanical and electric power. In addition solar and geothermal energies can be used directly for heating applications. The theoretically available potential and conversion technologies of these renewable energy resources are discussed briefly here.

1.5 Solar Energy

Sun is the source of all life on the earth. All forms of energy on the earth are derived from the sun. The sun is considered to be a sphere of intensely hot gaseous matter continuously generating heat by thermonuclear fusion reactions which convert hydrogen atoms to helium atoms. Fuelled by the thermonuclear fusion, the sun radiates energy into space at a rate of nearly 4×10^{26} watt. This energy is in the form of electromagnetic radiation with a wide range of wavelengths from short wavelength X-rays to long wavelength radio waves. But 99% of the solar radiation lies in the narrow band of visible spectrum between the ultraviolet and infrared.

Solar energy is a very large, inexhaustible source of energy. The power from the sun intercepted by the earth is approximately 1.8×10^{11} MW which is many thousand times larger than the present consumption rate on the earth of all commercial energy sources. Thus, in principle, solar energy could supply all the present and future energy needs of the world on a continuing basis. This makes it one of the most promising of the unconventional energy sources. The advantages of solar energy are environmentally clean source of energy, and freely available in adequate quantities in almost all parts of the world where people live.

The main problems associated with solar energy are :

- (i) **Dilute source of energy** : Even in the hottest regions on earth, the solar radiation flux available rarely exceeds 1 kW/m^2 and the total radiation over a day is at best about 7 kWh/m^2 . These are low values from the point of view of technological utilization. Consequently, large collecting areas are required in many applications and these result in excessive losses.
- (ii) **Availability varies widely with time** : Variation occurs at a specific location because of local weather conditions. Consequently, the energy collects when the sun is shining must be stored for use during periods when it is not available.
- (iii) The need for storage also adds significantly to the cost of any system.

1.5.1 Application of Solar Energy

- (i) Heating and cooling of buildings
- (ii) Solar water and air heating
- (iii) Salt production by evaporation of sea water
- (iv) Solar distillation
- (v) Solar drying of agricultural products

1.5.2 Basic Principle in Solar Thermal Energy Systems

In any collection device, the principle usually followed is to expose a dark surface to solar radiation so that the radiation is absorbed. A part of the absorbed radiation is then transferred to a fluid like air or water. When no optical concentration is done, the device in which the collection is achieved is called a flat-plate collector. When higher temperatures are required, it becomes necessary to concentrate the radiation. This is achieved using focussing or concentrating collectors. For low temperature applications i.e., below 100°C , flat plate-collector is used, while for higher temperatures concentrating collectors are used.

1.5.2.1 Flat Plate Collector

The flat plate collector mainly consist of a casing, absorber plate, transparent glass covers, insulating material and fluid passage tubes (Fig. 1.1).

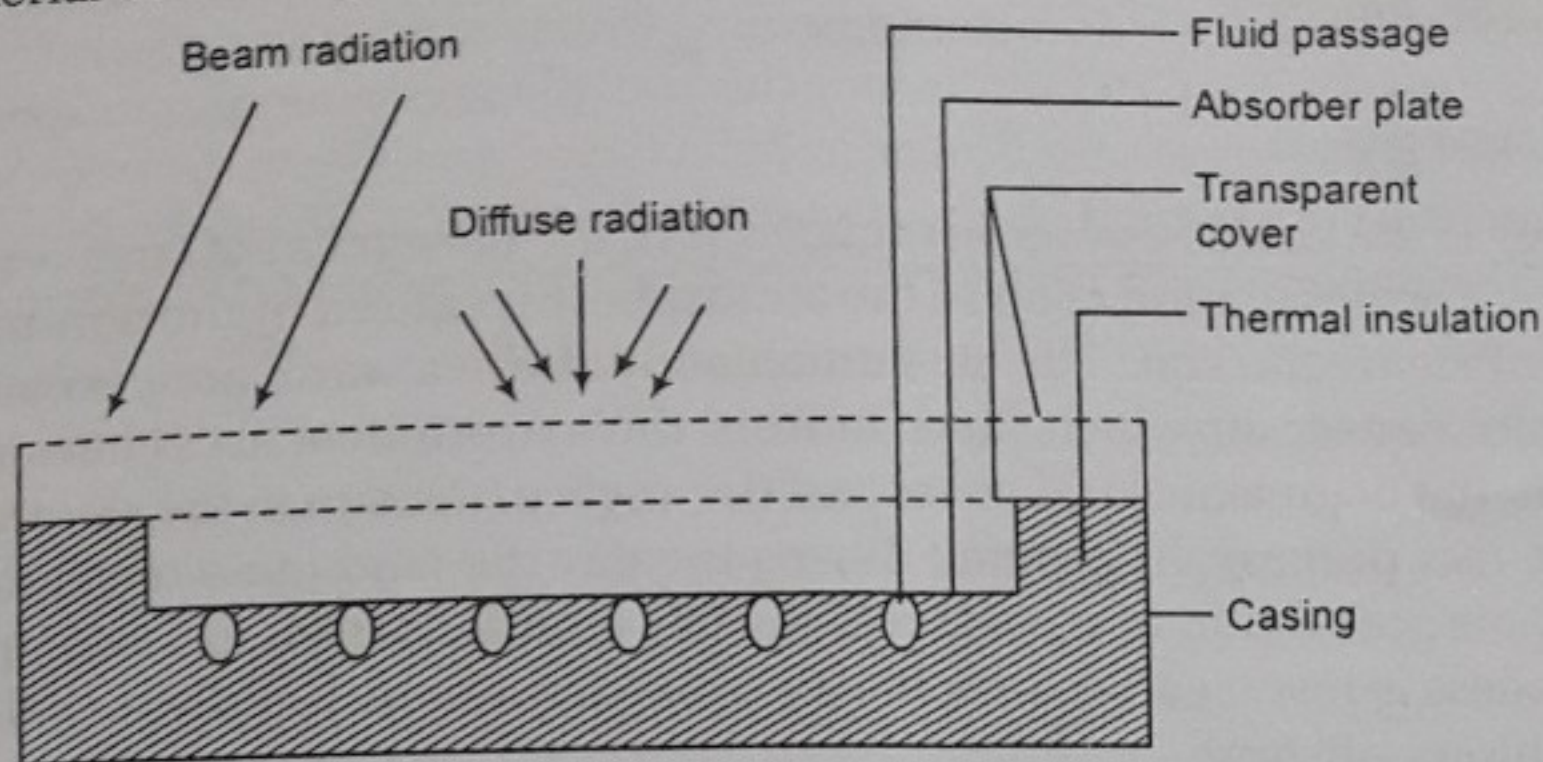


Figure 1.1 Schematic diagram of liquid flat-plate collector

When the collector is exposed to sun, the radiation (**Beam, diffuse radiation**) pass through transparent glass covers. The radiation is absorbed by absorber plate, which is coated with black absorber paint. The fluid tubes, which are connected to absorber plate, absorb the heat and transferred to the water passing through the tubes and gets heated. The hot water collected from all the tubes flow into a storage tank. Insulation (paddy husk, saw dust, glass wool) provided to the absorber plate to avoid loss of heat by conduction. Loss of heat by re-radiation is avoided by having good absorber coating. The convective loss is reduced by minimizing the air gap between the glass covers. The temperature attained is in the range of 40-100°C depending upon the intensity of solar radiation. A liquid flat-plate collector is usually held tilted in a fixed position on a supporting structure, facing north if located in the northern hemisphere.

1.5.2.2 Cylindrical Parabolic Concentrating Collector

In this type of collector (Fig. 1.2), the radiation coming on concentrator is diverted to absorber tube, which is covered with glass tube to avoid re-radiation loss.

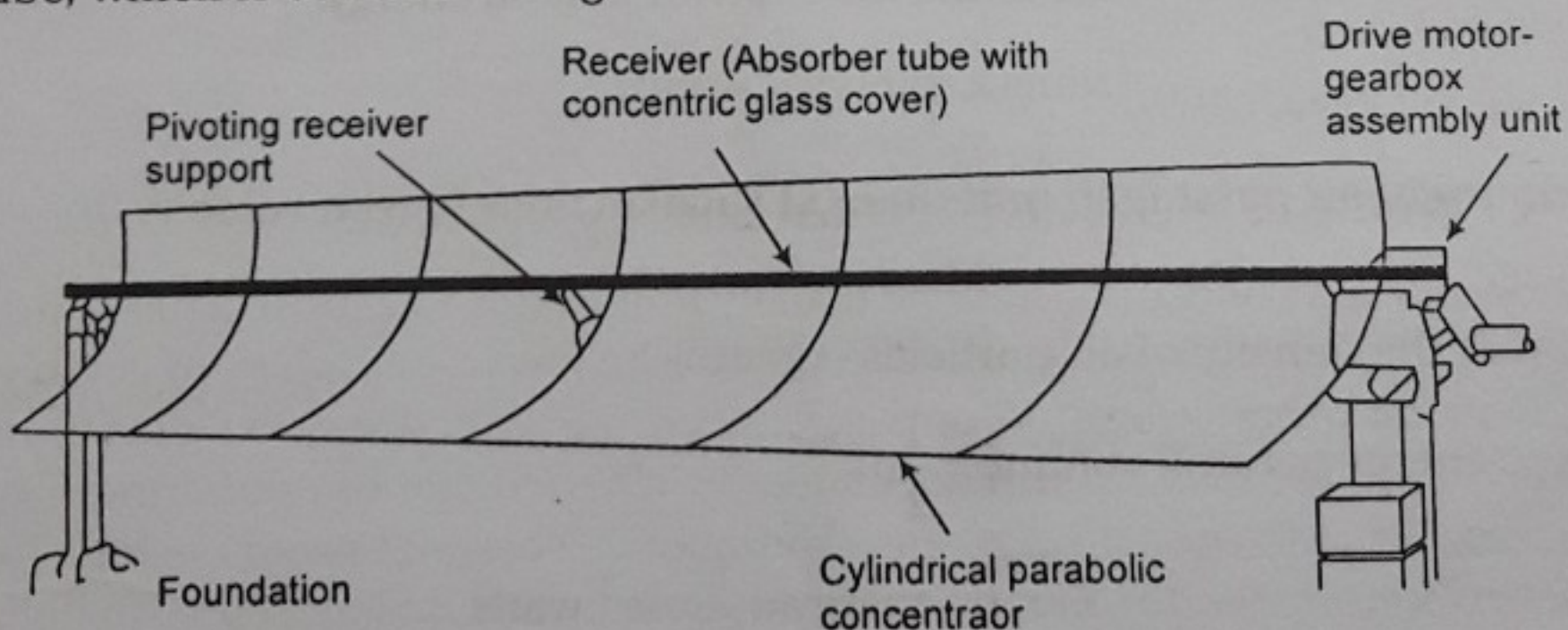


Figure 1.2 Schematic diagram of cylindrical parabolic concentrating collector

The heat from the absorber tube is transmitted to the fluid passing and gets heated up. The hot liquid flows out to a storage tank for further application. The direction of concentration is changed with the sun movement by providing single axis tracking. The temperature attained by the collector is $100-300^{\circ}\text{C}$.

1.6 Wind Energy

Wind energy is the kinetic energy associated with the movement of large masses of air over the earth surface. The circulation of air in the atmosphere is caused by the non-uniform heating of the earth's surface by the sun. The air immediately above a warm area expands, it is forced upwards by a cool denser air which flows in from the surrounding areas causing a wind. The nature of terrain, the degree of cloud cover and the angle of the sun in the sky are all the factors which influence this process. In general, during the day the land mass tends to heat up more rapidly than the air over water. In coastal regions this manifests itself in a strong onshore wind. At night this process is reversed because the air cools down more rapidly over the land and the wind therefore blows offshore.

The main planetary winds are caused in much the same way. The earth receives maximum amount of solar radiation at the equator and minimum at the poles. The cool surface air sweeps down from the poles forcing the warm air over the tropics to rise. But the direction of these massive air movements is affected by the rotation of the earth and the net effect is large counter-clockwise circulation of air around low pressure areas in the northern hemisphere, and clockwise circulation in the southern hemisphere. The strength and direction of these planetary winds change with season as the solar energy input varies. Despite the wind's intermittent nature, wind flow patterns at any particular place remains remarkably constant year by year. Average wind speeds are greater in hilly and coastal areas than they are well inland.

Wind speeds increase with height. They have been traditionally measured at a standard height 10 m, where they are found to be 20-25% greater than close to the surface. At a height of 60 m, they may be 30-60% higher because of the reduction in drag effect of the earth surface.

1.6.1 The Power in the Winds

Winds mills converts the kinetic energy of the wind into mechanical energy. The total power of the wind stream is equal to the time rate of kinetic energy.

$$\text{Kinetic energy} = \frac{1}{2}mv^2 \quad \dots(1.1)$$

The amount of air passing in unit time through an area A , with velocity, v

$$\text{Mass, } m = \rho Av \quad \dots(1.2)$$

where, ' ρ ' is the density of air particles (kg/m^3)

$$\therefore \text{Kinetic energy per unit volume} = \frac{1}{2}\rho v^2 \quad \dots(1.3)$$

$$\text{Kinetic energy} = \frac{1}{2}\rho Av^3 \text{ watts} \quad \dots(1.4)$$

$$\text{Total Power (P}_{\text{total}}) = \frac{1}{2} \rho A v^3 \text{ watts}$$

where, P_{total} is the total power in the wind stream and put in equation (1.5).
A wind turbine converts the kinetic energy of the wind to rotary mechanical energy. Blowing wind spins the blades on a wind turbine. When the wind blows, a pocket of low pressure air forms on the downwind side of the blade. The low pressure air pockets then pull the blade towards it, causing the rotor to turn. This is called lift. The force of the lift is actually much stronger than the wind's force against the front side of the blade, which is called drag. The combination of lift and drag causes the rotor to spin like a propeller. The rotor of the turbine are attached to a hub that is mounted on a turning shaft.

The shaft goes through a gear transmission box where the turning speed is increased. The transmission is attached to a high speed shaft which turns a generator that makes electricity. The electricity thus obtained can depending on the end user's need, either alternating or direct. Direct current can be generated by driving brushless D.C. generator attached to the windmill's rotor. Wind turbine can be used as stand alone applications.

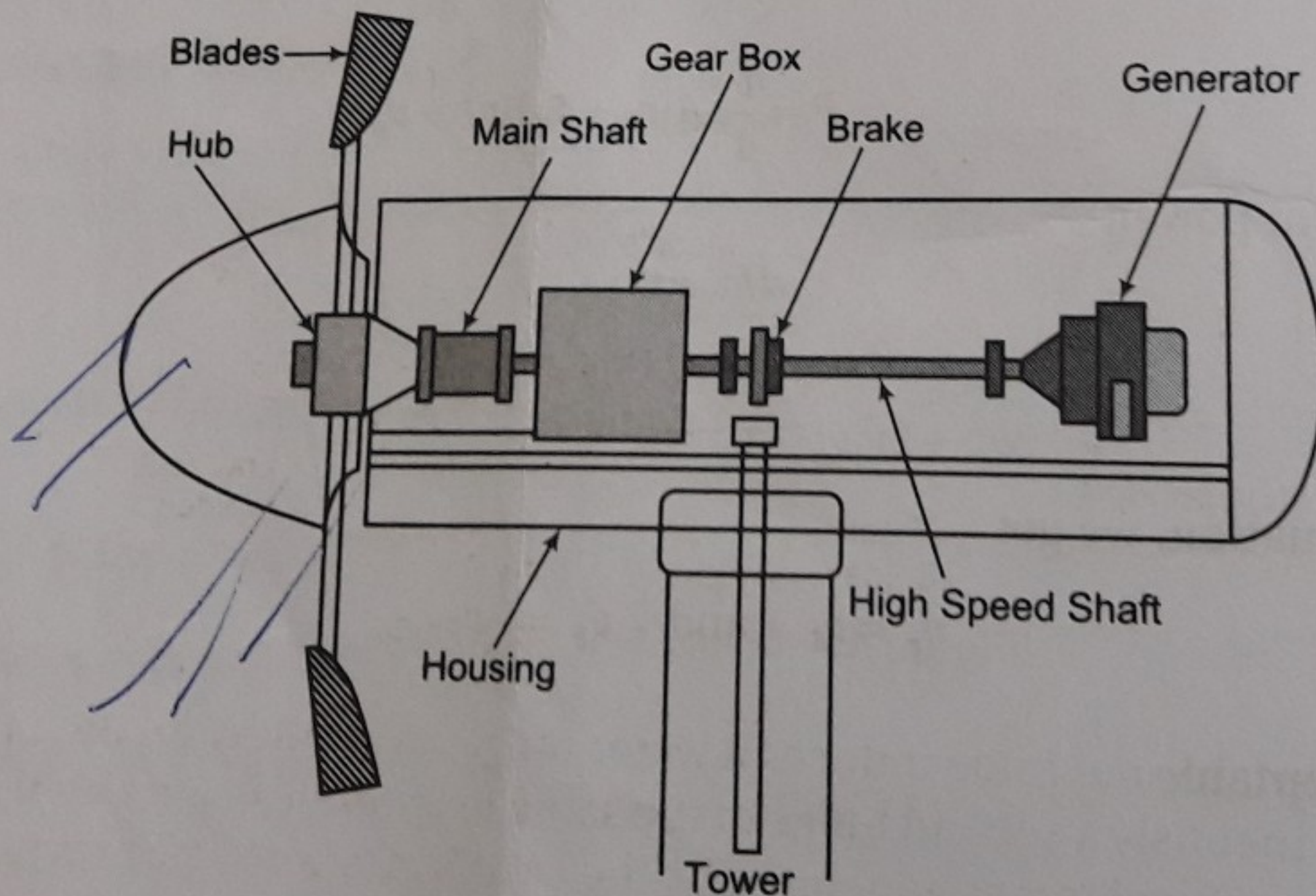


Figure 1.3 Wind Turbine.

1.6.2 Advantages and Disadvantages of Wind Energy

1.6.2.1 Advantages of Wind Energy

1. It is renewable source of energy.
2. It is non-polluting and no adverse influence on the environment
3. No fuel and transportation is required.
4. The cost of electricity under low production is comparatively low.

1.6.2.2 Disadvantages of Wind Energy

1. The available wind energy is dilute and fluctuating in nature.
2. Unlike water energy, wind energy requires storage capacity because of its irregularity.
3. Wind energy operating machines are noisy in operation.

There are two main types of windmill. They are :

- (i) **Horizontal axis type** : Multiblade type windmill, sail type windmill, propeller type windmill.
- (ii) **Vertical axis type** : Savonius type windmill, Darrieus type windmill.

1.7 Tidal Energy

The basic concept of tidal energy is to capture the potential energy of the raising and lowering tides. The gravitational forces exerted on the earth by the sun and the moon cause a displacement of ocean water. As the earth rotates the surface of the ocean water alternately rises and falls at a particular location. Although the rise and fall of tides are mainly due to the gravitational pull by the moon as it circles round the earth, the nature of tide depends on the position of the sun with respect to moon and the earth, and the rotation of the earth itself about its polar axis.

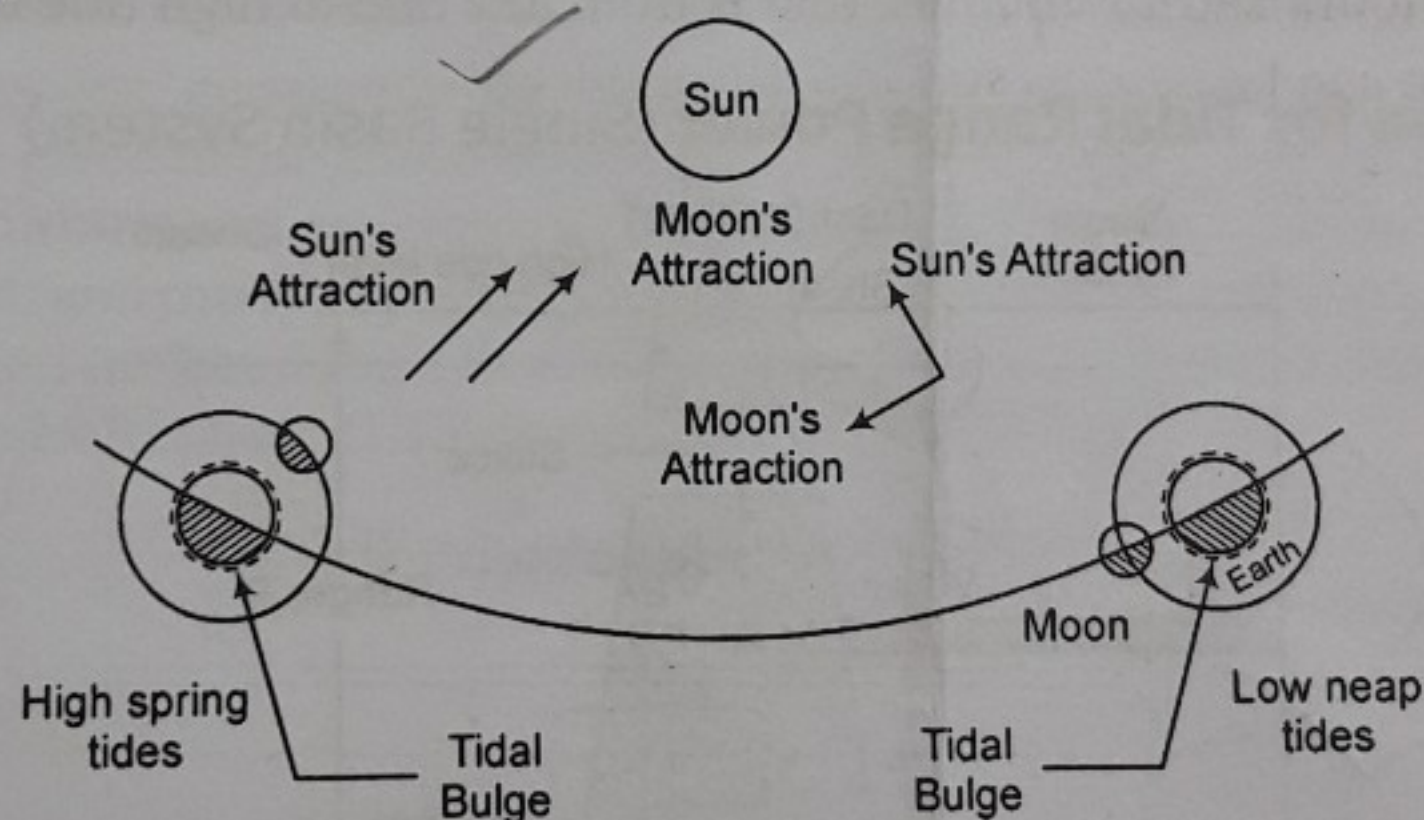


Figure 1.4 Spring and neap tides

The moon's gravitational force induces a water bulge, called tide, on the near side surface of the earth, simultaneously a second bulge of the same magnitude occurs on the far side of the earth as result of centrifugal forces generated by the earth's rotation. The gravitational forces of

1.7.1 Basic Principle of Tidal Energy Conversion

When the tides occur, the rise of water increases its potential energy. This increase in the potential energy of water can be converted into mechanical energy by the water turbines. The main components of tidal power plant are : The dam, sluice ways from the basin to the sea and vice versa, turbine with generators etc. The sluice ways or, gate and water turbines are installed in the barrages. The function of dam (barrages) is to form a barrier between the sea and the basin or, between one basin to other basin in case of multiple basins. When the high tides occur in the ocean, the flood gates are opened so as to allow the flow of water into the pool thereby increasing the potential energy of water in the pool. When the low tides occur the sluice gates are close and the water with high potential energy flows from the pool through the water turbines to generate power. The pool is filled and emptied during one tidal cycle that extends for a period of 12 hours and 25 minutes, that is from low tide to high tide to low tide.

1.7.2 Expression for Tidal Range Power (Single Basin System)

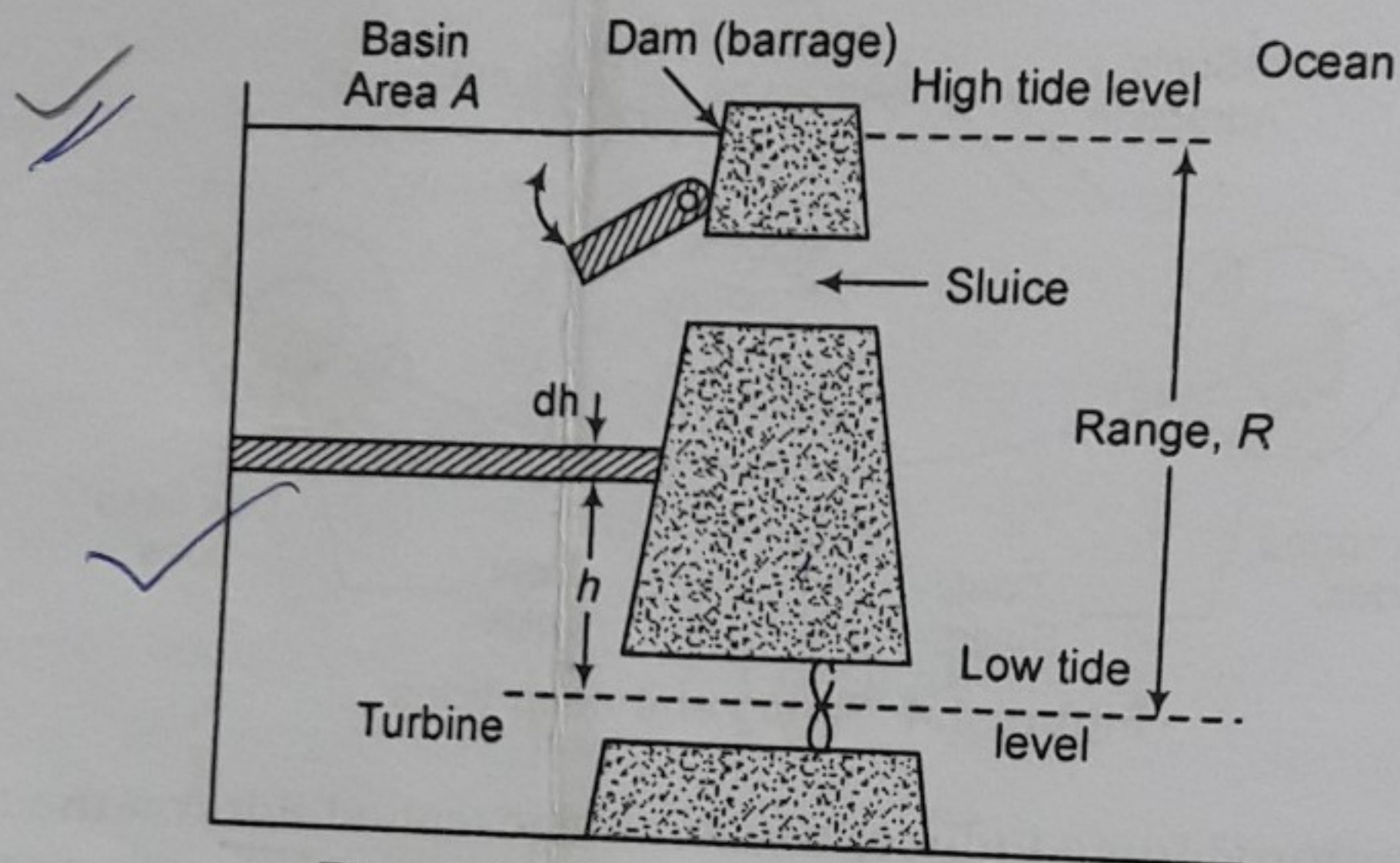


Figure 1.5 Power generation from tides.

✓ 1.7.3 Advantages and Disadvantages of Tidal Power

✓ 1.7.3.1 Advantages of Tidal Power

- (i) Water is a free resources and is inexhaustible.
- (ii) No Pollution.
- (iii) Free from the problems of uprooting the people and disturbing the ecology balance.

- (iv) No extra submerging of land is involved.
- (v) It produces 24 hrs a day all year around.
- (vi) Peak output coincides with peak energy demand.
- (vii) It is 80% efficient where as coal is only 30%.

1.7.3.2 Disadvantages of Tidal Power

- (i) Variability in output power due to the variation in the tidal range
- (ii) The tidal ranges is highly variable and thus the turbines have to work on a wide range of head variation. This affects the efficiency of the plant.
- (iii) Initial cost is more and tidal technology is not yet economically feasible
- (iv) Technology is still being developed.

1.8 Ocean Thermal Energy (OTE)

Ocean thermal energy conversion is based on tapping energy potential created by temperature difference between sun-warmed surface water and deep polar feed bottom currents to generate electricity. The solar radiation emitted by the sun is absorbed by the sea water and its temperature is raised. The upper layers of the sea water acts as a storage device for heat. Then a heat source at a high temperature and heat source at low temperature and this temperature gradient or, difference can be utilized in prime mover (turbine) that can convert part of the heat into mechanical energy and then into electrical energy. The residual heat is discharged to the sink.

In fact, the ocean is the world's largest solar collector. Absorption of solar energy in water takes place according to **lambert's law of absorption** that states each layer of equal thickness absorbs the same fraction of light that passes through it. Expressing mathematically

$$-\frac{dI_{(x)}}{dx} = kI_{(x)} \quad \text{or} \quad \boxed{I_{(x)} = I_0 e^{-kx}}$$

where I_0 and $I_{(x)}$ are intensities of radiation at the surface (i.e., $x = 0$) and at depth x from the surface respectively. The factor k is known as extinction coefficient or, absorption coefficient. The value of k for very salty sea water is 0.5 m^{-1} (for clear fresh water, $k = 0.05 \text{ m}^{-1}$). At this rate, more than 95% absorption occurs within a depth of 6 m. Thus, almost all the absorption occurs very close to the surface raising its temperature, while the deep water at about 100 m or, so remains cool and unaffected by the solar radiation. Due to this warmer water stays at top providing a huge heat source while the cold water stays at a depth providing a huge heat sink. Both reservoirs are maintained actually by solar radiation.

The maximum possible efficiency of a heat engine working between two temperature limits can not be more than that of a Carnot cycle operating between the same temperature limits. In the range of temperature of warm water (T_h) in the upper surface layer and cold water (T_c) in the depth of the tropical ocean, the Carnot cycle efficiency is given by

$$\eta = 1 - \frac{T_c}{T_h} = \frac{T_h - T_c}{T_h} = \frac{\Delta T}{T_h}$$

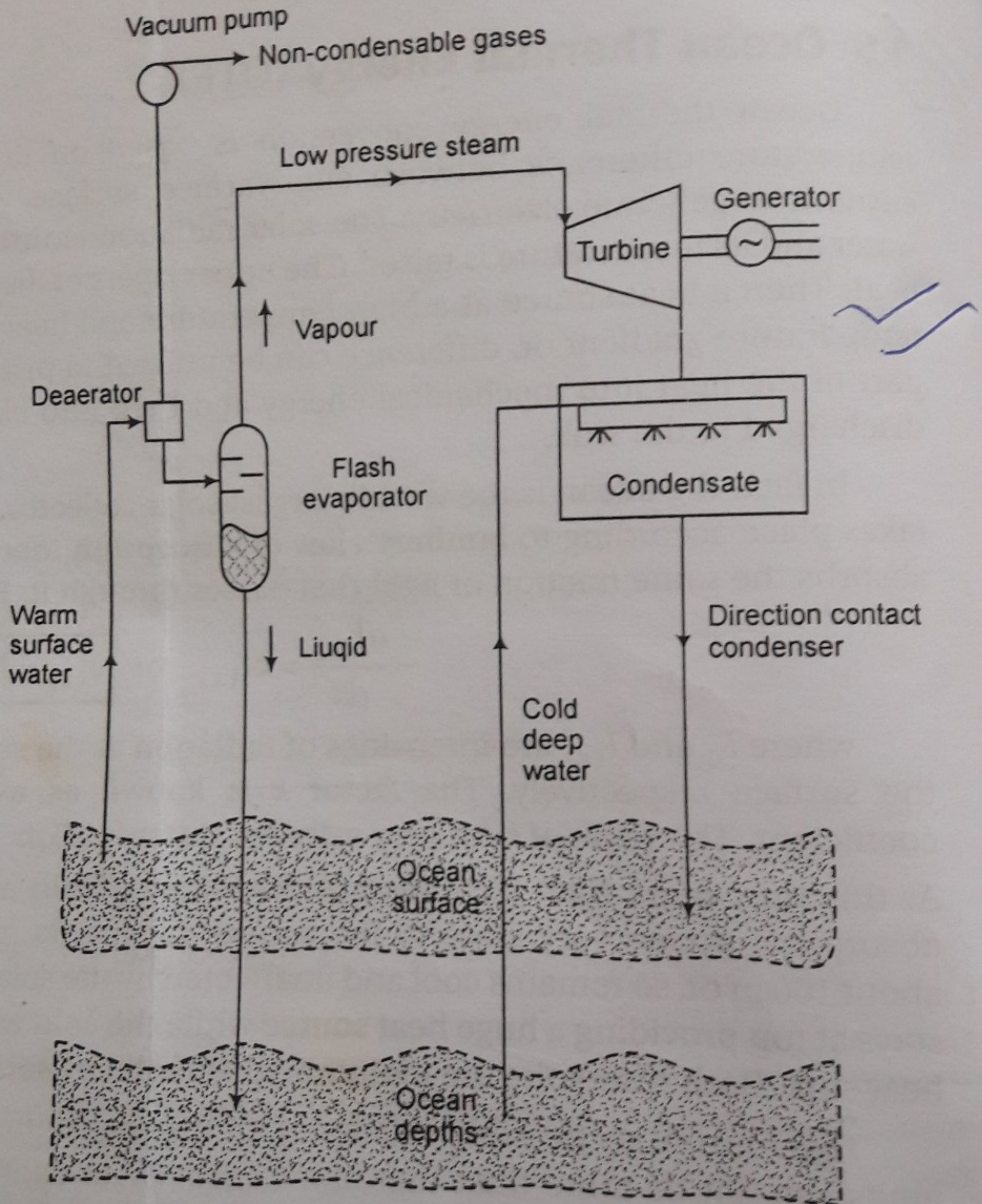


Figure 1.6 Open cycle OTEC plant

in as Anderson cycle) plant, warm surface water is used to
working fluid

1.8.1 Advantages and Disadvantages of OTE

1.8.1.1 Advantages of OTE

- (i) Renewable energy resources.
- (ii) The resource supplies steady power without fluctuations.
- (iii) Seasonally independent resources.
- (iv) Only marginal improvements in existing machinery (i.e., heat exchanger, turbine etc) required for use.
- (v) It also has the ability to create some useful byproducts such as desalinated water and nutrients for mariculture.

1.8.1.2 Disadvantages of OTE

- (i) High installation cost.
- (ii) Low thermal efficiency (around 3-4%).
- (iii) Bio-fouling means deposition and growths of microorganism on the cooling water side of the heat exchanger. It reduces the heat transfer efficiency.
- (iv) Problems of power transmission to the shore submarine cables are needed for this.

1.9 Geothermal Energy

Geothermal energy is the thermal energy trapped beneath and within the solid crust of the earth. This energy manifests through steam, hot water and hot rock, and is released naturally in the form of geysers, hot springs and volcanic eruptions.

The earth is considered as a ball of molten rock surrounded by a thin crust of hard crystalline rock. The crust is only about 30 kilometer thick, its outer surface is cool, but the temperature increase rapidly with depth. In regions where the crust is especially thin, or, where there is high volcanic or, seismic activity, thermal gradients can rise by over 80°C for each 1000 penetrated. In geologically stable areas, temperature gradients are much lower averaging between 25° and 35° per 1000 m depth. There are four types of geothermal resources: hydrothermal, geopressured, hot dry rock and magma. Only hydrothermal is commercially exploited.

Steam geysers and hot water springs occur where the rock is sufficiently permeable or, fractured to permit water to find its way to the surface. At the locations where natural geyser and spring do not occur, the geothermal energy can be mined by drilling down into a hot underground aquifer—a layer of porous rock containing hot water. The heat from the hot impermeable rocks can be extracted by fracturing rocks at depth and then pumping water through the fissure, either the steam or, the hot water is obtained through the drilled hole or, fissures.

Hydrothermal convective systems are again sub-classified as :

- (i) Vapour dominated or, dry steam fields
- (ii) Liquid dominated system or, wet steam fields, and hot water fields.

(i) **Dry steam system** : In this system, the liquid particles and suspended solids are eliminated by using centrifugal separators. The condensate coming out of condenser was fed

back to the hot field with the help of pump. The supply of steam diminishes due to depletion of thermal energy of that region or, insufficient supply of ground water. Therefore, to keep up the steam supply, new wells must be drilled. The dry steam availability is rare.

(ii) **Liquid dominated system** : Fig. 1.9 shows a closed system geothermal power plant using hot water as the main heat source and a secondary fluid driving turbogenerator. In this system, water is pumped to the surface by a deep-well pump. To ensure that no steam is produced and no dissolved gases escape from the water, the water is kept above saturation pressure throughout the process.

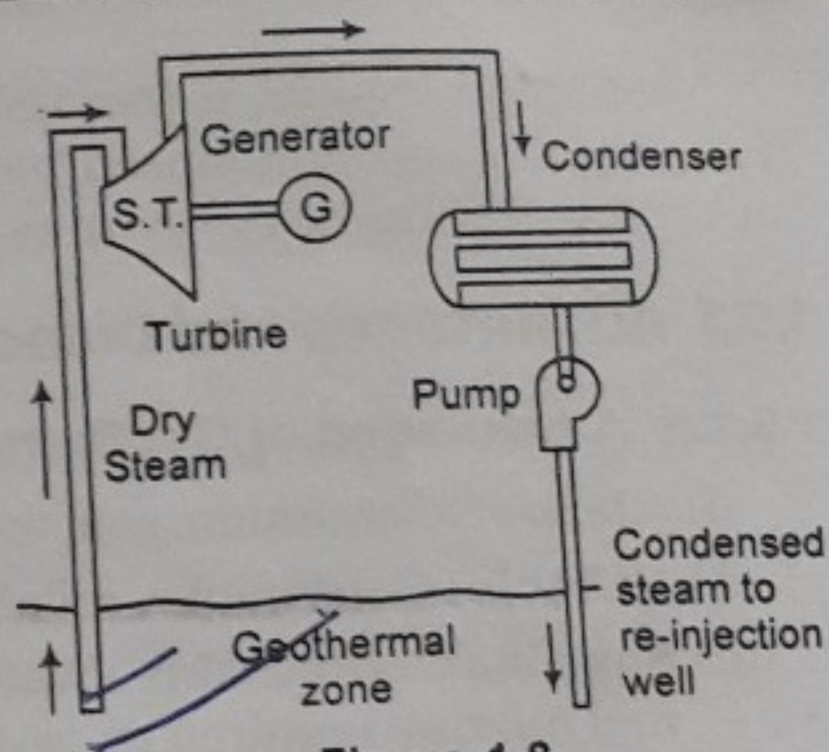


Figure 1.8

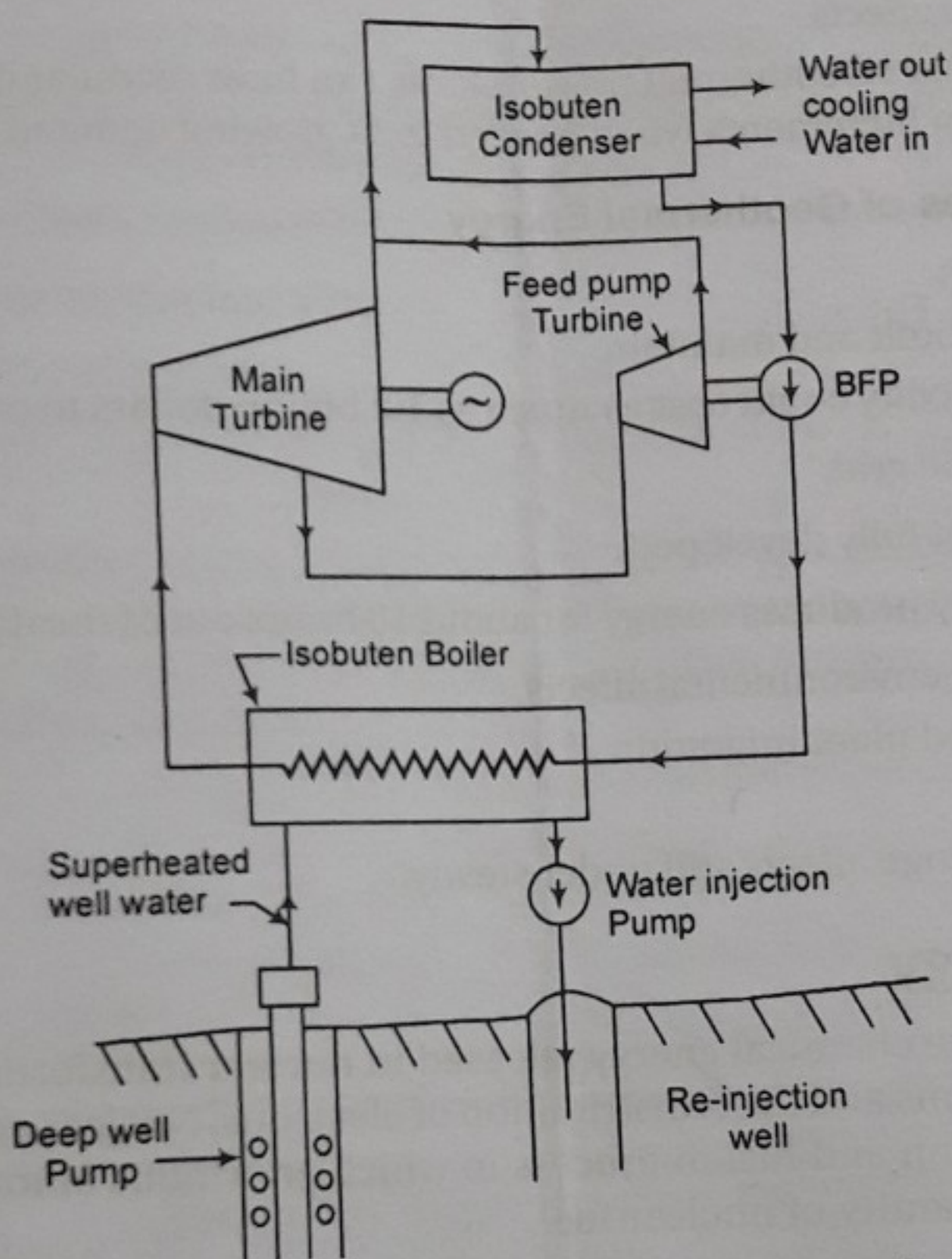


Figure 1.9 Closed system geothermal power plant

The water gives up its heat in a heat exchanger (boiler) to heat a liquid (isobutane) and superheat it. The cooler water is then injected into a well nearby carrying all dissolved gases and most of the solids in solution with it. The water thus injected maintains the underground reservoir pressure zone and very likely may be reheated by the hot porous zone. Here the water becomes a means of transporting heat from the hot rocks beneath to the power plant, in a continuous cycle.

Isobutane vapour, expanding through a turbine, yields its energy to drive a generator. The exhaust gas is condensed back to the liquid in a water cooled condensation and pumped back to the isobutane boiler to repeat the cycle.

1.9.1 Advantages and Disadvantages of Geothermal Energy

1.9.1.1 Advantages of Geothermal Energy

- (i) **Clean** : No emission, safe to use.
- (ii) **Reliable** : Continuous, reliable base load power.
- (iii) **Sustainable/Reusable** : water can be recycled back into the earth and reused. No other fuel mixture required to create electricity.
- (iv) **Land conservative** : No major land requirements can be integrated into the local area with no adverse effects.
- (v) **Flexible/modular** : Geothermal power plants can have modular designs, with additional units installed in increments when needed to fit growing demand for electricity.

1.9.1.2 Disadvantages of Geothermal Energy

- (i) Presently costly :
 - ⇒ Expensive to built and maintain
 - ⇒ A 1085 MW facility could cost as much as 1.2 billion dollars to construct and run
- (ii) Connection to the grid.
- (iii) Technology is not fully developed.
- (iv) Barrage style only produces energy for about 10 hours out of the day.
- (v) Barrage style has environmental affects.
 - ⇒ Such as fish and plant migration.
 - ⇒ Silt deposits.
 - ⇒ Local tides change affects still under steady.

1.10 Nuclear Energy

Nuclear energy is the chemical energy released in nuclear transformations resulting from the rearrangement of atoms and the redistribution of electrons. Nuclear energy can be obtained by both through the fission and fusion process in which enormous amounts of energy will be released from small a quantity of nuclear fuel.

Natural uranium consists of two isotopes, namely, uranium 235 (U_{235}) and Uranium 238 (U_{238}) in the ratio of 1 : 139. The less abundant isotope uranium 235 can be more easily split by bombardment with neutrons than uranium 238.

It is found that plutonium 239 and uranium 233 (the former obtained from uranium 238 and the latter obtained from uranium 232, both by absorption of neutrons) are also capable of being easily split by neutrons and thus enormous amount of energy can be released.

Uranium 235, plutonium 239 and uranium 233 are therefore called fissile materials. These are used as nuclear fuels for generating power. On the other hand, uranium 238 and uranium 232 are called fertile materials. These can be converted inside a nuclear reactor into fissile materials.

Enriched uranium, which is nothing but the natural uranium in which the concentration of the isotope uranium 235 has been increased, is also used as a fuel in nuclear power plants.

1.10.1 Advantages and Disadvantages of Nuclear Energy

1.10.1.1 Advantages of Nuclear Energy

- ✓ 1. Almost zero emission (very low greenhouse gas emission).
2. Can be sited almost anywhere unlike oil which is mostly imported.
3. A small amount of matter creates a large amount of energy.
- ✓ 4. A lot of energy is generated from a single power plant.
5. Nuclear waste may be reprocessed if legally allowed.
6. Breeder reaction create more usable fuel than they use.

1.10.1.2 Disadvantages of Nuclear Energy

- ✓ 1. expensive to build and maintain.
- ✓ 2. waste products are dangerous and need to be carefully stored for a long periods of time.
- ✓ 3. Safety problem.

Fission is the process that occurs when a neutron collides with the nucleus of certain of the heavy atoms, causing the original nucleus to split into two or more unequal fragments which carry off most of the energy of fission as kinetic energy. This process is accompanied by the emission of neutron and gamma rays.

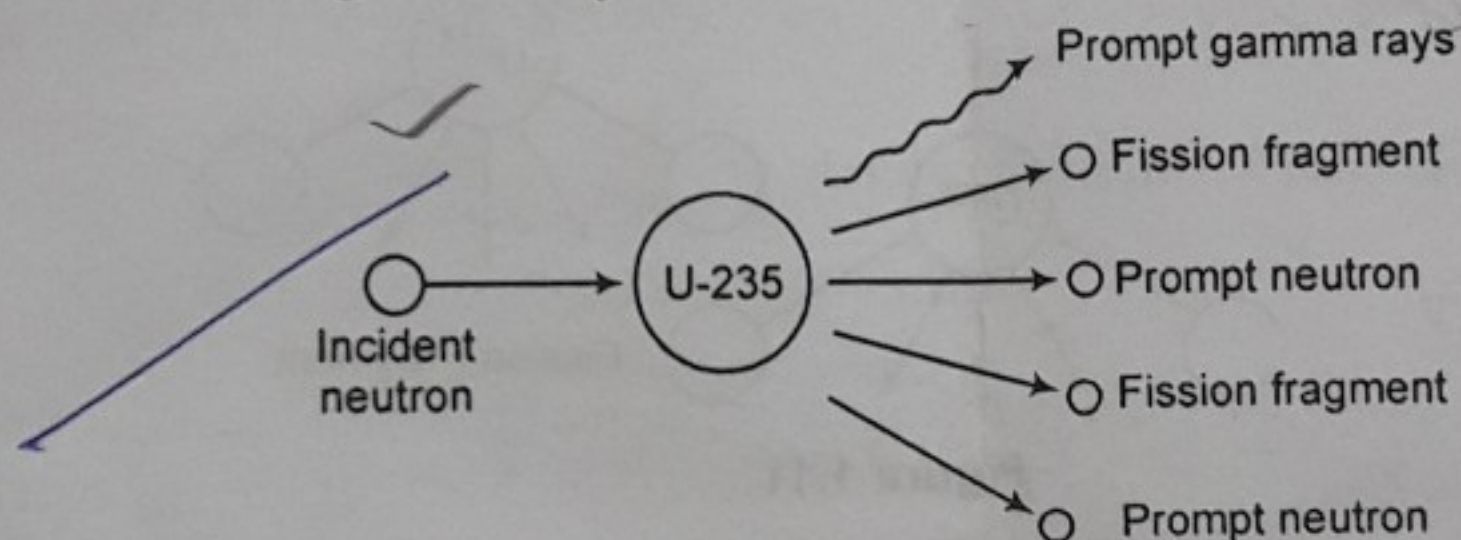


Figure 1.10 Fission of uranium 235.

Incident neutron upon colliding with U^{235} nucleus, causes fission to take place, resulting in the production of fission fragments, prompt neutrons and prompt gamma rays.

- A self sustaining series of reactions, in particular those of nuclear fission in which the particles released by one nucleus trigger the fission of at least as many further nuclei.
- A reaction that simulates its own reaction.

Example : Nuclear reaction occurs when neutrons released from a split atom such as a uranium 235(U^{235}) nucleus go on to fission other nuclei. This reaction produces additional

neutrons that cause more fissions, which release still more neutrons to cause even more fissions, which release even more neutrons and so on.

A chain reaction is that process in which the number of neutrons keeps on multiplying rapidly (in geometrical progression) during fission till whole of the fissionable material is disintegrated.

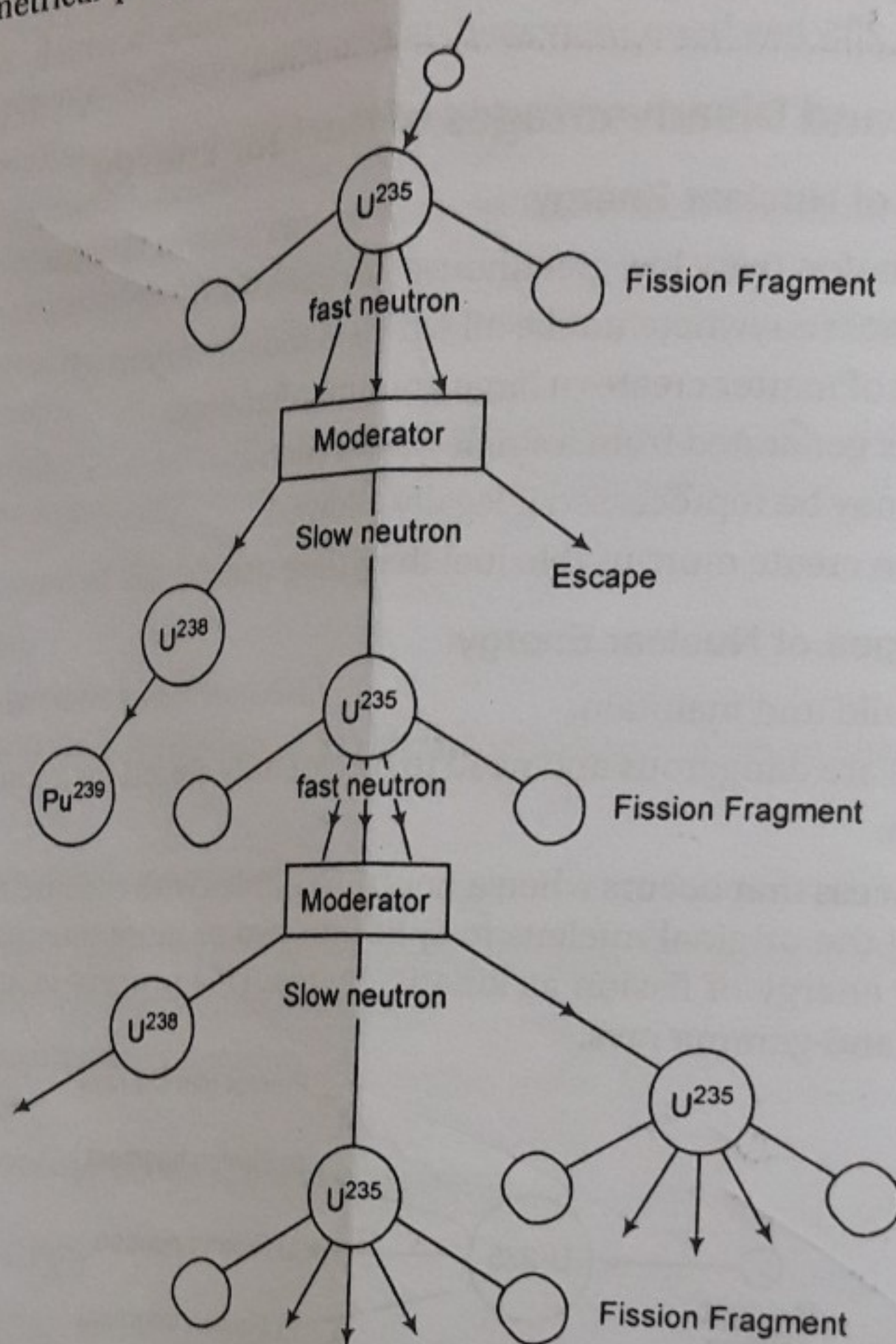


Figure 1.11

When the nucleus ${}_{92}U^{235}$ is bombarded by a neutron, the neutron is absorbed in the uranium nucleus resulting in a nucleus of higher mass number which is highly excited and agitated and it splits up within a very short interval of time. The fragments produced in the breakup of the heavy nucleus are isotopes of elements located in the middle part of the periodic table where the nuclear binding energy per neutron is greater than in heavy nuclei. Thus, there is considerable release of energy in fission process. At the same time, on the average 2 to 3 neutrons are emitted per fission reaction. These neutrons can strike other nuclei of ${}_{92}U^{235}$ and cause fission. In this manner a continuous chain reaction will result. The result of collision between the nucleus and the neutron depends upon its speed. If the neutron passes close to a

U^{235} nucleus at a slow speed, the short range attractive forces will be able to pull it into the nucleus than if it is travelling at high speed. The neutrons emitted as a result of chain reaction travel at a very high speed, e.g., about 1.5×10^9 cm/sec and are termed as "Fast neutrons". Therefore to sustain a chain reaction, the high speed should be reduced corresponding to the speed of molecules in a gas at normal temperature and pressure, i.e., 2.2×10^5 cm/sec. Such a neutrons have an energy of about 0.0027 eV. This is achieved by passing the fast neutrons through a moderator material and colliding with molecules.

In such a reaction, the number of neutrons will go on multiplying as there is no control over them. This will result in infinite release of energy. But we need energy at a uniform and controlled rate. So, the chain reaction must be a controlled one. In nuclear reactor, control rods absorb some of the neutrons so that chain reaction is controlled and energy is released at a steady and uniform rate.

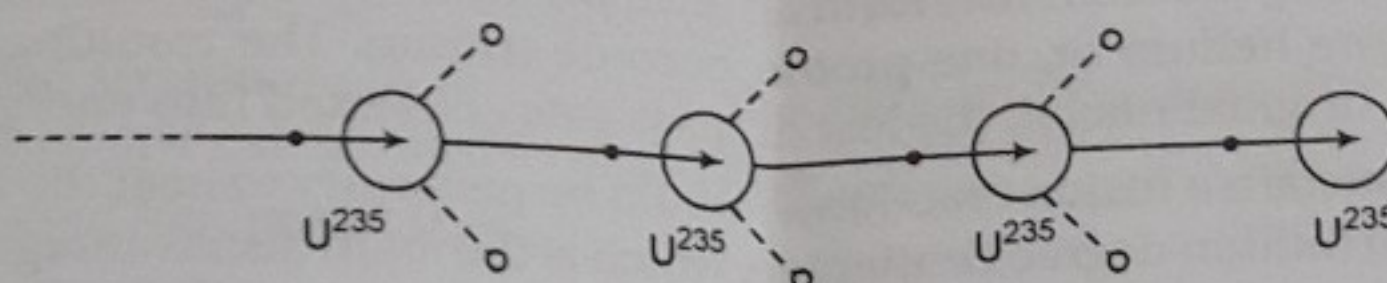


Figure 1.12

The fission reaction is possible only if the excitation energy exceeds a threshold value known as "critical energy". The excitation energy is supplied by the absorbed neutron e.g if a neutron is absorbed by U^{235} , an energy of 6.5 MeV is supplied to the nucleus.

1.10.2 Reproduction Factor

Reproduction Factor can be maintained Unity or More to Sustain the Chain Reaction:

1. If the ${}_{92}U^{235}$ % in uranium content is increased from 0.7% to 10% then the number of neutron captured by ${}_{92}U^{238}$ are reduced and probability of fission is increased. In this process, fast and slow neutron are used to maintain the chain reaction.

The enrichment of ${}_{92}U^{235}$ is not possible by chemical means as ${}_{92}U^{235}$ can not separated from ${}_{92}U^{238}$ by chemical process. The gaseous diffusion process is used to enrich to ${}_{92}U^{235}$ which is costly

2. In this method, the fast neutrons are thermalised as mentioned earlier and the thermalised neutrons are capable to maintain the chains reaction with the natural uranium having 0.7% fissible ${}_{92}U^{235}$. In this method, the danger of absorption of emitted neutrons by ${}_{92}U^{238}$ is reduced avoiding the resonance range of ${}_{92}U^{238}$

The four factor formula : The four factor formula also known as fermi's four factor formula is used in nuclear engineering to determine the multiplication of a nuclear chain reaction in an infinite medium. The formula is

$$k_{\infty} = \eta f p \epsilon$$

where, η is re-production factor and it is defined as the number of fission neutrons produced per absorption in the fuel.

where f is the thermal utilization factor and it is defined as probability that a neutron that gets absorbed does so in the fluid material.
 p is the resonance escape probability and it is defined as fraction of fission neutrons that manage to slow down from fission to thermal energies without being absorbed.
 ϵ is the fast fission factor and it is ratio of total no. of fission neutrons and no. of fission neutrons from just thermal fissions.

Nuclear fusion is the process of combining or, fusing two lighter nuclei into a stable and heavier nuclei. In this case also large amount of energy is released because mass of the product nucleus is less than the masses of the two nuclei which are fused.

In the fusion process, when light of nuclei such as deuterium and tritium – the two forms of hydrogen, are combined the excess binding energy is released. When two nuclei of deuterium are forced together they momentarily form an unstable nucleus which immediately release one neutron and become helium or, one proton become tritium. The resulting nucleus has less mass than the two original nuclei, the lost of mass gets converted into energy.

To ignite and sustain a fusion reaction, it would be necessary to heat the fuel to a temperature in excess of 100 million degree centigrade, which is the main disadvantage in this process.

A reactor based on deuterium tritium fusion would release 80 percent of its energy in the very fast moving neutrons. These neutrons could heat a jacket of liquid lithium eventually producing unable electricity from a conventional steam powered generator, Neutrons would also cause fission in some lithium nuclei producing tritium fuel for the basic fusion reaction with deuterium.

1.10.3 Comparison of Fission and Fusion Processes

The comparison between "Fission" and "Fusion" processes is given below

Fission	Fusion
1. When heavy unstable nucleon is bombarded with neutrons, the nucleus splits into fragments of equal mass and energy is released.	1. Some light elements fuse together with the release of energy.
2. About one thousandths of the mass is converted into energy.	2. It is possible to have four thousandths of mass converted into energy.
3. Nuclear reaction residual problem is great.	3. Residual problem is much less.
4. Amount of radioactive material in a fission reactor is high.	4. A possible advantage is that the total amount of radioactive material in a working fusion reactor is likely to be very much less than that in a fission reactor.
5. Because of higher radio active material, health hazards is high in case of accidents.	5. Because of lesser radio active material, health hazards is much less.
6. It is possible to construct self sustained fission reactors and have positive energy release.	6. It is extremely difficult to construct controlled fusion reactors.
7. Manageable temperatures are obtained.	7. Needs unmanageable temperatures like 30 million degrees for fusion process to occur.
8. Raw fissionable material is not available in plenty	8. Reserves of deuterium, the fusion element, is available in great quantity.

A nuclear reactor is an apparatus in which nuclear fission is produced in the form of a controlled self sustaining chain reaction.

The nuclear reactors are classified on the following basis :

1. On the basis of Neutron Energy

- (a) **Fast Reactors** : In these reactors, the fission is effected by fast neutrons without any use of moderators.
- (b) **Thermal Reactors** : In these reactors, the fast neutrons are slowed with the use of moderators. The slow neutrons are absorbed by the fissionable fuel and chain reaction is maintained. The moderator is most essential component in these reactors.

2. On the basis of Fuel Used

- (a) **Natural Fuel** : In this reactor, the natural uranium is used as fuel and generally heavy water or graphite is used as moderator.
- (b) **Enriched Uranium** : In this reactor, the uranium used contains 5 to 10% U^{235} and ordinary water can be used as moderator.

3. On the basis of Moderator Used

- (a) Water moderated
- (b) Heavy water moderated
- (c) Graphite moderated
- (d) Beryllium moderated.

4. On the basis of Coolant Used

- (a) Water cooled reactors (ordinary or heavy)
- (b) Gas cooled reactors
- (c) Liquid metal cooled reactors.
- (d) Organic liquid cooled reactors.

Few of the reactors which are commonly used for power generation are discussed below.

1.10.4 Pressurised Water Reactor (PWR)

In its simplest form, a pressurised water reactor is a light water cooled and moderated reactor. It uses enriched uranium as fuel. The pressurising tank included in the circuit maintains the constant pressure in the circuit throughout the load range. Electric heating coil in the pressurised boils the water to form the steam which is collected in the dome as shown in figure and pressurises the entire coolant circuit before starting reactor. To reduce the pressure, water spray is used to condense the steam.

The fuel which is generally used is UO_2 . The water in the primary circuit gets heated by absorbing the fission energy in the reactor core and same energy is given in the heat exchanger to generate the steam. The water carry out of the heat exchanger circulated by the pump to maintain the pressure in the circuit in the range of 100 to 130 bar. The pressure in the primary circuit should be high so that the boiling of water takes place at higher pressure. A pressuring tank keeps the water at about 100 kg/cm^2 so that, it will not boil.

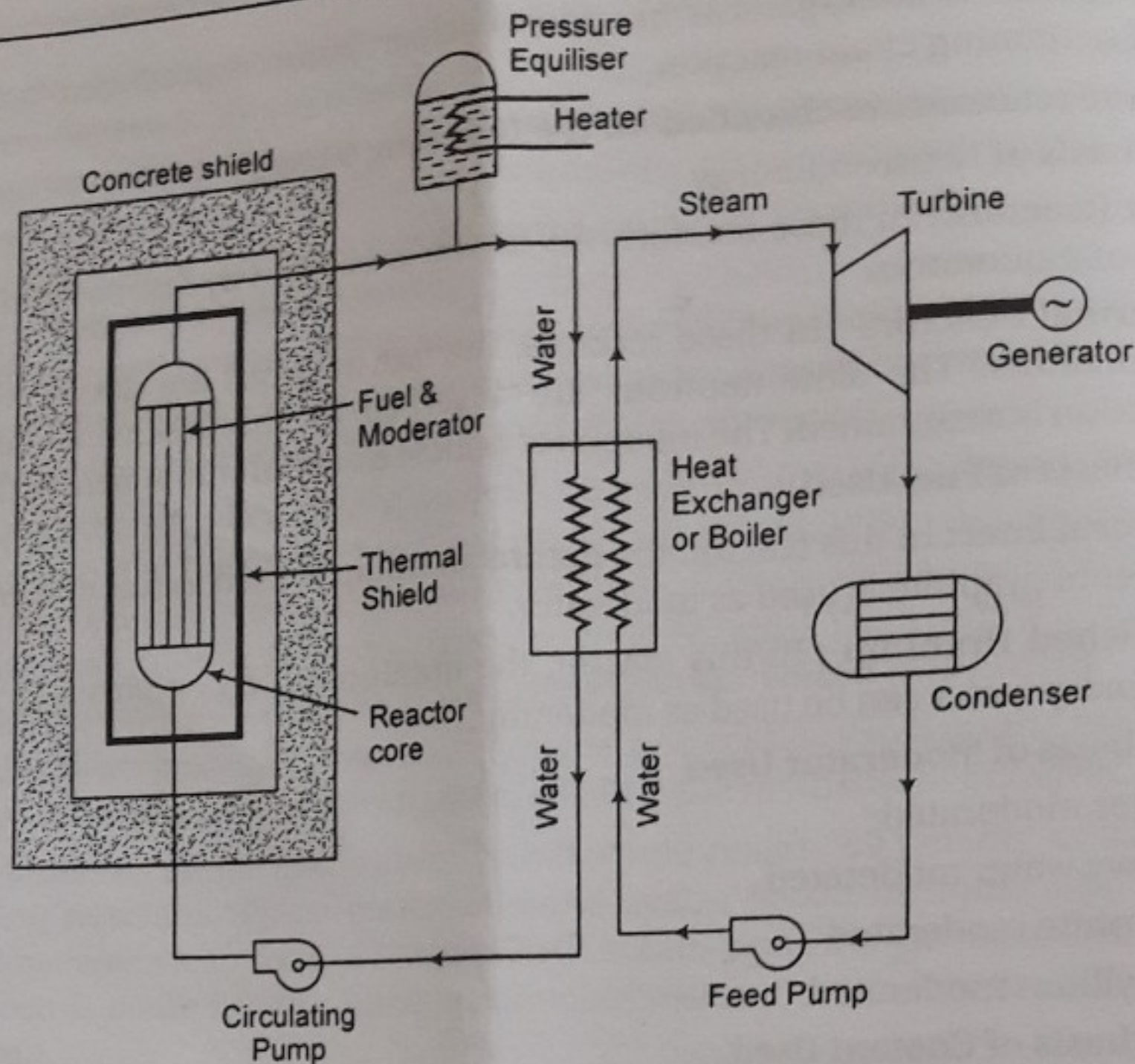


Figure 1.13 Outline of pressurised water reactor nuclear power plant.

Electric heating coils in the pressurised boiler boil some of the water to form steam that collects in the dome. As more steam is forced into the dome by boiling, its pressure rises and pressurises the entire circuit. A PWR can produce only saturated steam.

Advantages

1. Water used in reactor (as coolant, moderator and reflector) is cheap and easily available.
2. The reactor is compact and power density is high.
3. Fission products remain contained in the reactor and are not circulated.
4. A small number of control rods is required.

Disadvantages

1. Capital cost is high as high primary circuit requires strong pressure vessel.
2. Fuel suffers radiation damage and, therefore its reprocessing is difficult.
3. Severe corrosion problem.
4. It is imperative to shut down the reactor for fuel charring which requires a couple of month's time.

1.10.5 Boiling Water Reactor (BWR)

In this type of reactor, enriched uranium is used as fuel and water is used as coolant, moderator and reactor like *PWR* except the steam is generated in the reactor itself instead of separate steam boiler.

Inside the *BWR* vessel, a steam water mixture is produced when very pure water (reactor coolant) moves upward through the core absorbing heat. The major difference in the operation as a *BWR* from other is the steam void formation in the core. The steam water mixture leaves the top of the core and enters the two stages of moisture separation, where water droplets are removed before the steam is allowed to enter the steam line. The recirculation pumps and set pumps allow the operator to vary coolant flow through core and change reactor power.

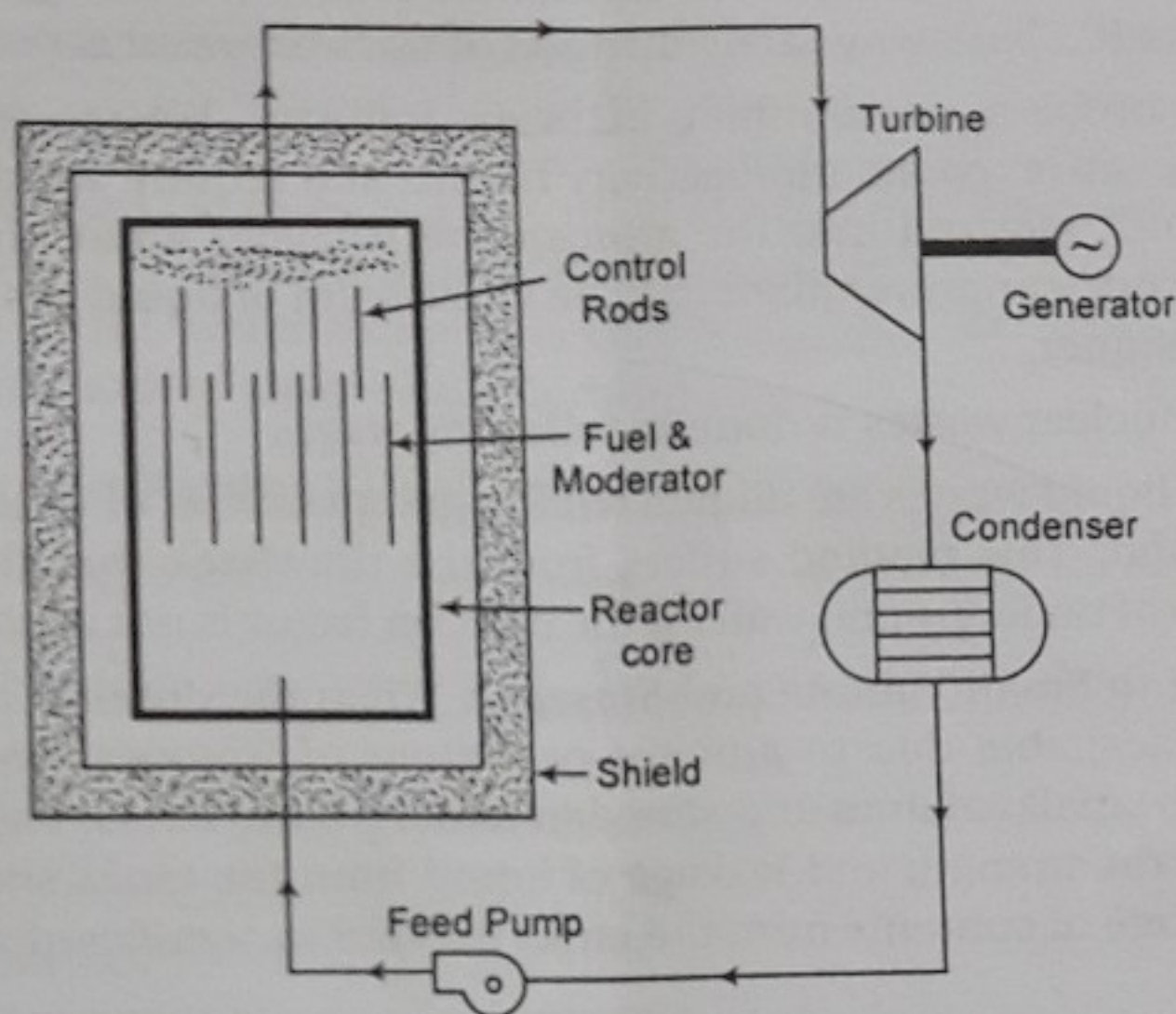


Figure 1.14 Boiling water reactor power plant with single circuit.

Advantages of *BWR*

- (i) The pressure inside the reactor vessel is considerable smaller than *PWR* as water is allowed to boil inside the reactor. Therefore, the reactor vessel can be much lighter than *PWR* and reduces the cost of pressure vessel considerably
- (ii) This reactor does not require steam generator, pressuring circulating pump and connecting piping. Therefore, the cost is further reduced.
- (iii) A boiling water reactor is more stable than other.
- (iv) Thermal efficiency of the reactor plant (30%) is considerably higher than *PWR* plant (20-22%).